

PRACTICAL NO. 11

Aim

Preparation of Case Study Report on “Break-Even Point Calculation of a Successful Enterprise” using Amul as a case study.

1. Introduction

Break-Even Point (BEP) is an important financial concept used to determine the level of sales at which total revenue becomes equal to total cost. At the break-even point, the enterprise has neither profit nor loss.

Break-even analysis helps an entrepreneur understand the relationship between fixed cost, variable cost, selling price and sales volume. It can be used for production planning, pricing decisions, cost control and estimating the minimum quantity that must be sold to avoid losses.

This report studies Amul as a successful enterprise and presents an illustrative break-even point calculation for a representative Amul milk product. Since enterprise-level internal fixed-cost and product-level contribution data are not publicly available in the reference material, the numerical calculation uses clearly stated assumptions for academic purposes rather than claiming to be Amul's actual internal BEP.

2. Selected Enterprise

For understanding break-even analysis, the following successful enterprise is considered:

Enterprise Name: Amul

Nature of Business: Dairy products, milk procurement, processing and marketing.

Type of Enterprise: Large cooperative dairy enterprise/brand network.

Basic Characteristics:

- Large milk procurement network
- Dairy processing and manufacturing
- Wide product portfolio
- Strong distribution network
- Recognised consumer brand
- Cooperative business structure
- Large-scale production
- Focus on affordable dairy products

3. Major Factors Affecting Break-Even Point

The following factors are considered while calculating and analysing the break-even point:

Factor	Major Effect on BEP
Fixed Cost	Higher fixed cost generally increases the break-even quantity
Variable Cost	Higher variable cost reduces contribution per unit and increases BEP
Selling Price	Higher selling price generally increases contribution and reduces BEP
Contribution	Difference between selling price and variable cost
Sales Volume	Sales above BEP generate operating profit under the simplified model
Product Mix	Different products can have different prices, costs and contribution margins
Capacity	Production capacity affects the feasible sales volume
Cost Control	Efficient cost management can lower the break-even level

4. Amul – About the Enterprise

Amul is the brand associated with the Gujarat Cooperative Milk Marketing Federation (GCMMF), which markets dairy products produced through a cooperative network. The Amul model connects milk producers with processing and marketing activities and has developed into a major dairy brand in India.

Amul's product range includes milk, butter, cheese, ghee, milk powder, ice cream and other dairy products. Its large-scale distribution and established brand presence make it suitable for studying the financial concept of break-even analysis.

Business Model

- Milk procurement through cooperative members
- Processing and value addition
- Large-scale dairy manufacturing
- Brand-based marketing
- Wide distribution network
- Multiple dairy product categories
- Retail and institutional sales

5. Meaning of Break-Even Point

The Break-Even Point is the level of sales at which total revenue equals total fixed and variable costs.

At BEP:

Total Revenue = Total Cost

Profit = Zero

Loss = Zero

The basic break-even quantity formula is:

$$\text{BEP (Units)} = \text{Fixed Cost} \div (\text{Selling Price per Unit} - \text{Variable Cost per Unit})$$

The amount (Selling Price – Variable Cost) is called Contribution per Unit.

6. Assumptions for the Case Study

The following assumptions are made only to demonstrate the break-even calculation for a representative packaged milk product. They are not presented as Amul's actual confidential cost structure.

Particular	Assumed Value
Product	Representative 1-litre packaged milk unit
Selling Price per Unit	₹60
Variable Cost per Unit	₹42
Contribution per Unit	₹18
Annual Fixed Cost	₹18,00,000
Expected Sales Volume	1,50,000 units

The assumptions are simplified so that the break-even calculation can be clearly understood.

7. Break-Even Point Calculation

Step 1: Calculate Contribution per Unit

Contribution per Unit = Selling Price – Variable Cost

= ₹60 – ₹42

= ₹18 per unit

Step 2: Calculate Break-Even Quantity

BEP = Fixed Cost ÷ Contribution per Unit

= ₹18,00,000 ÷ ₹18

= 1,00,000 units

Therefore, the representative product must sell approximately 1,00,000 units to reach the break-even point.

Step 3: Calculate Break-Even Sales Value

BEP Sales = BEP Units × Selling Price

= 1,00,000 × ₹60

= ₹60,00,000

Particular	Result
Selling Price per Unit	₹60
Variable Cost per Unit	₹42
Contribution per Unit	₹18
Fixed Cost	₹18,00,000
Break-Even Quantity	1,00,000 units
Break-Even Sales	₹60,00,000

8. Profit Analysis Above Break-Even Point

Assuming the representative enterprise sells 1,50,000 units:

$$\text{Sales Revenue} = 1,50,000 \times ₹60 = ₹90,00,000$$

$$\text{Total Variable Cost} = 1,50,000 \times ₹42 = ₹63,00,000$$

$$\text{Contribution} = ₹90,00,000 - ₹63,00,000 = ₹27,00,000$$

$$\text{Estimated Profit} = \text{Contribution} - \text{Fixed Cost}$$

$$= ₹27,00,000 - ₹18,00,000$$

$$= ₹9,00,000$$

Thus, under the stated assumptions, sales of 1,50,000 units would generate an illustrative operating profit of ₹9,00,000.

9. Margin of Safety

Margin of Safety indicates how much actual or expected sales can fall before the enterprise reaches the break-even level.

$$\text{Margin of Safety (Units)} = \text{Actual Sales} - \text{Break-Even Sales}$$

$$= 1,50,000 - 1,00,000$$

$$= 50,000 \text{ units}$$

$$\text{Margin of Safety (\%)} = 50,000 \div 1,50,000 \times 100$$

$$= 33.33\%$$

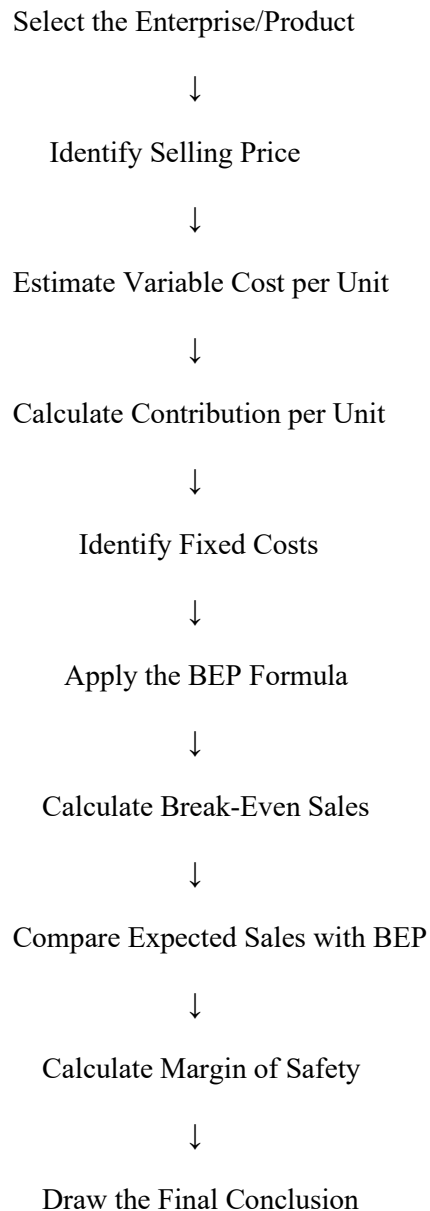
Therefore, under the assumptions, sales could decline by approximately 33.33% from the expected level before reaching the break-even quantity.

10. Break-Even Analysis of the Selected Enterprise

Parameter	Observation
Selling Price	₹60 per unit (assumed)
Variable Cost	₹42 per unit (assumed)
Contribution	₹18 per unit (calculated)
Fixed Cost	₹18,00,000 (assumed)
Break-Even Quantity	1,00,000 units
Break-Even Sales	₹60,00,000
Expected Sales	1,50,000 units
Estimated Profit	₹9,00,000
Margin of Safety	50,000 units / 33.33%

The calculation shows that once sales exceed 1,00,000 units, the simplified model begins generating operating profit. Actual enterprise-level BEP would depend on the complete cost structure, product mix, capacity utilisation, taxes and other financial factors.

11. Suggested Steps for Break-Even Point Calculation



12. Key Findings

- Break-Even Point identifies the sales level at which total revenue equals total cost.
- Contribution per unit is an important component of break-even analysis.
- Higher fixed costs generally increase the break-even quantity.
- Higher variable costs reduce contribution and generally increase the break-even quantity.
- Higher selling price can increase contribution and reduce the break-even quantity, provided demand remains adequate.
- Under the assumptions used in this case study, the break-even quantity is 1,00,000 units.
- Expected sales of 1,50,000 units are above the break-even level by 50,000 units.
- The calculated margin of safety is 33.33% under the stated assumptions.
- Actual enterprise-level break-even analysis requires detailed internal cost and product-mix data.

13. Conclusion

The case study of Amul demonstrates the practical importance of Break-Even Point analysis for an enterprise. Break-even analysis helps management understand the minimum sales volume required to recover fixed and variable costs and provides a useful basis for production, pricing and sales planning.

Using the stated academic assumptions for a representative packaged milk product, the break-even point is calculated at 1,00,000 units, corresponding to break-even sales of ₹60,00,000. At an assumed sales volume of 1,50,000 units, the simplified calculation gives an illustrative operating profit of ₹9,00,000.

Therefore, break-even analysis is a useful decision-making tool for successful enterprises. However, actual financial decisions should use verified internal cost data, product-wise contribution margins, product mix, capacity constraints and current market conditions.

15. References

MSBTE, Entrepreneurship Development and Startups, Course Code 315002.

Amul / Gujarat Cooperative Milk Marketing Federation (GCMMF), official company information.

Amul, official product and business information.

Basic principles of Cost and Management Accounting – Break-Even Analysis.